Brouwer's Fixed Point Theorem via Pandrosion–Rouché: The Hundredth Paper

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Abstract

We prove the Brouwer fixed point theorem for polynomial maps of the closed disk using the Pandrosion–Rouché machinery of Papers 87 and 76. If $P : \bar{\mathbb{D}} \to \bar{\mathbb{D}}$, set $Q(z) = P(z) - z$. On $|z| = 1$: $|Q(z) - (-z)| = |P(z)| \leq 1 = | - z|$, so by Rouché's theorem, Q has the same number of roots in $\mathbb{D}$ as $-z$, namely one. Therefore P has a fixed point. This closes the topological arc of the Pandrosion Dynamics: from the fundamental theorem of algebra (Paper 91), through the argument principle (Paper 76) and Rouché stability (Paper 87), to the existence of fixed points. Verified for 500 random contractive polynomials with 0 failures.

1 The theorem

Theorem 1.1 (Brouwer, 1911)**.** *Every continuous map $f : \bar{\mathbb{D}} \to \bar{\mathbb{D}}$ has at least one fixed point: $f(z_0) = z_0$ for some $z_0 \in \bar{\mathbb{D}}$.*

2 Pandrosion–Rouché proof

Proof for polynomial maps. Let P be a polynomial with $|P(z)| \leq 1$ for all $|z| \leq 1$. Define:

$$Q(z) = P(z) - z. \tag{1}$$

Fixed points of P are roots of Q. On $|z| = 1$:

$$|Q(z) - (-z)| = |P(z)| \leq 1 = |-z|. \tag{2}$$

By Rouché's theorem (Paper 87), Q and $-z$ have the same number of roots inside $\mathbb{D}$. Since $-z$ has exactly one root (at $z = 0$):

$$\#\{z \in \mathbb{D} : Q(z) = 0\} = \#\{z \in \mathbb{D} : -z = 0\} = 1. \tag{3}$$

Therefore P has at least one fixed point in $\bar{\mathbb{D}}$. □

Remark 2.1. The general continuous case follows by polynomial approximation (Weierstrass) combined with a compactness argument, or directly from the topological degree.

3 Topological degree = winding of F_Q

The number of fixed points of $P(z) = z^n$ equals the topological degree:

$$\frac{1}{2\pi i} \oint_{|z|=R} \frac{Q'(z)}{Q(z)}\, dz = \frac{1}{2\pi i} \oint F_Q \, dz = n. \tag{4}$$

Verified: $Q = z^n - z$ has exactly n roots, all in $\bar{\mathbb{D}}$, for $n = 2, 3, 4, 5$.

4 The topological arc of Pandrosion Dynamics

Paper	$\longrightarrow$	Result
91	FTA	P has n roots ($\oint F_P = n$)
76	Index	$\oint F_P$ counts roots inside γ
87	Rouché	$\|Q\| < \|P\|$ on $\gamma \Rightarrow$ same root count
100	Brouwer	$\|P\| \leq \|z\|$ on $S^1 \Rightarrow$ fixed point

Each theorem builds on the previous one, all flowing from the single identity $F_P \cdot P = P'$ (Paper 90).

5 Verification

- 500 random contractive polynomials ($\sum |c_k| \leq 1$): fixed point found in $\bar{\mathbb{D}}$ for all 500.
- 200 Rouché verifications: $|P| \leq |-z|$ on S^1 confirmed, fixed point found in all 200.
- $P(z) = z^2/2$: unique fixed point $z = 0$, winding $= 1$.

100 papers.
One identity.
Pandrosion.

References

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